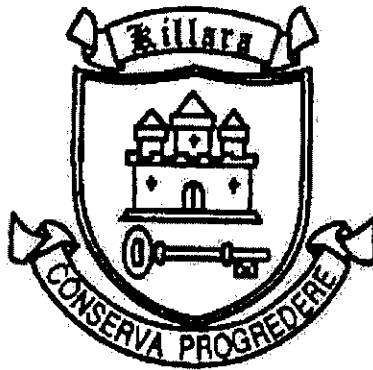


# KILLARA HIGH SCHOOL



**TERM 2 2009**

**TASK 3**

## **EXTENSION 1 MATHEMATICS EXAMINATION YEAR 12**

***Instructions:***

Time allowed: 70 minutes

Approved calculators may be used.

Show ALL working clearly.

Marks may not be awarded for careless or badly arranged work.

Begin each question on a NEW page

**Question 1 (15 marks)** (Use a NEW page)

- (a) (i) Find the largest possible domain of the function  $f(x) = (x-2)^2 + 5$  for which  $f(x)$  has an inverse function  $g(x)$ . **1**

- (ii) Find  $g(x)$  and hence sketch the graphs of  $f(x)$  and  $g(x)$  on the same set of axes. **3**

- (b). For the function  $f(x) = 3 \sin^{-1}(4x-1)$

- (i) State the domain and range of  $f(x)$ . **2**

- (ii) Hence sketch the graph of  $f(x)$  showing the major features. **1**

- (c) Differentiate  $y = x \tan^{-1} x$  and hence evaluate the exact value of  $\int_0^1 \tan^{-1} x \, dx$  **3**

- (d) Evaluate:

- (i)  $\int_0^{\frac{1}{2}} \frac{dx}{\sqrt{1-4x^2}}$  **3**

- (ii) Use the substitution  $u = \sin x$ , to evaluate  $\int_0^{\frac{\pi}{2}} \frac{\cos x \, dx}{1 + \sin^2 x}$  **2**

**Question 2 on next page**

**Question 2 (10 marks)** (Use a NEW page)

- (a) A particle moves in a straight line with an acceleration given by

$$\frac{d^2x}{dt^2} = 9(x - 2)$$

where  $x$  is the displacement in metres from an origin  $O$  after  $t$  seconds. Initially, the particle is 4 metres to the right of the origin  $O$  and has  $v = -6 \text{ ms}^{-1}$ .

- (i) Show that  $v^2 = 9(x - 2)^2$  **2**
- (ii) Find an expression for  $v$  and hence find  $x$  as a function of  $t$ . **2**
- (iii) Explain whether the velocity of the particle is ever zero. **2**

- (b) On a particular Saturday, the low-tide mark on a wharf was 1.4 m above sea bed and high tide 4 m above sea bed. The low-tide occurs at 9.05 a.m. and high-tide at 3.25 p.m. **4**

The top of the wharf is 4.3 m above seabed.

A barge tied to the wharf has its flat top 1.4 m above water level. Assuming that the rise and fall of the tide is a simple harmonic motion, calculate correct to the nearest minute, the time between low-tide and high-tide at which the top of the barge was exactly level with the top of the wharf.

*(Draw a diagram to represent the information)*

**Question 3 on next page**

**Question 3 (9 marks)** (Use a NEW page)

- (a) The volume,  $V \text{ m}^3$ , of usable wood in a tree of radius  $R$  metres can be modelled using the formula

$$\log_e V = 3 \log_e (2R) - 0.81$$

- (i) Using the approximation  $e^{0.81} = 2.25$ , show that the formula

$$\log_e V = 3 \log_e (2R) - 0.81 \text{ can be expressed as } V = \frac{32R^3}{9}.$$

- (ii) The radius of the tree is increasing at a rate of 0.002 metre per year.  
At what rate is the usable volume of the wood in the tree increasing when the radius of the tree is 1.2 metre?  
(Answer in  $\text{m}^3/\text{year}$  to 2 significant years)

- (b) An ice cube tray is filled with water at a temperature of  $18^\circ\text{C}$  and is placed in a freezer that has a constant temperature of  $-19^\circ\text{C}$ .

The cooling rate is proportional to the difference between the temperature of the freezer and the temperature of the water,  $T$ .

That is,  $T$  satisfies the equations

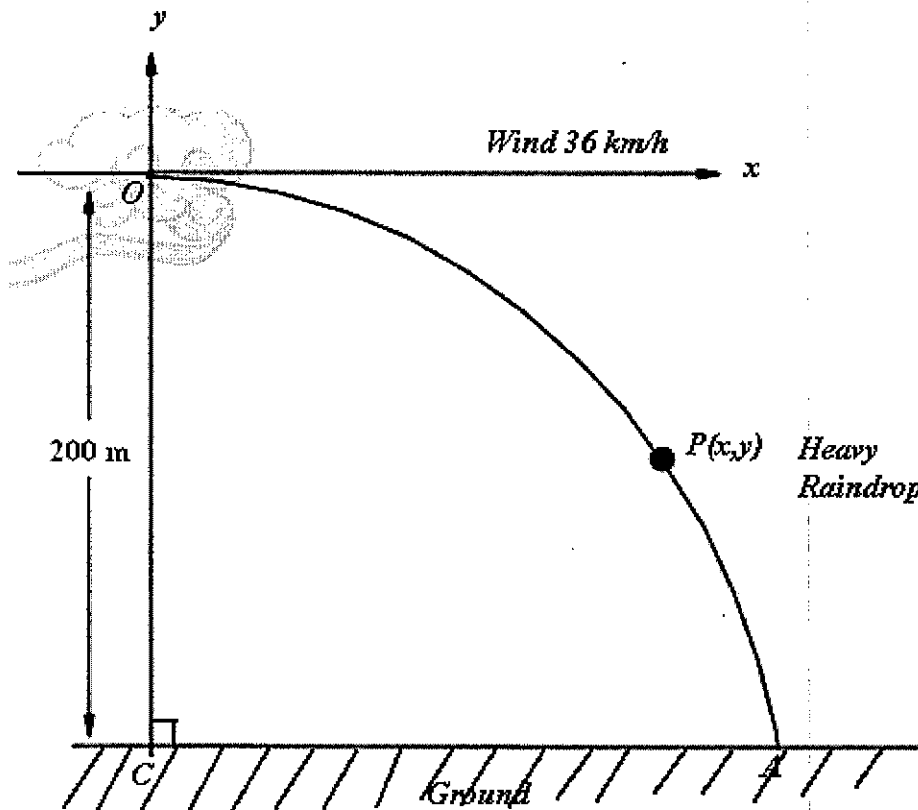
$$\frac{dT}{dt} = -k(T + 19) \text{ and } T = -19 + Ae^{-kt}$$

- (i) Show that  $A = 37$ .
- (ii) After 5 minutes in the freezer the temperature of the water is  $3^\circ\text{C}$ .  
Find the time for the water to reach  $-18.9^\circ\text{C}$ .  
Answer to the nearest minute.

**Question 4 continues on next page**

**Question 4 (8 marks)** (Use a NEW page)

A steady wind is blowing with a speed of 36km/h. From clouds moving horizontally with the wind, heavy raindrops fall to the ground 200 metres below.  
(Assume acceleration due to gravity  $g = 10\text{ms}^{-2}$ )



- (i) Derive the equations of motion. **2**
- (ii) Find the time taken for the drop to reach the horizontal ground. **1**
- (iii) Find the speed and angle at which a drop hits the horizontal ground. **3**
- (iv) At what angle does a drop hit the ground when the wind speed is doubled? **2**

**End of paper**

### 2009 Ext 1 Task 3 Marking guide lines

(a) (i)  $y = f(x) = (x-2)^2 + 5$

vertex is (2, 5)

$\therefore$  the domain of  $f(x)$  is  $x \geq 2$  for which  $f(x)$  has an inverse function **(1 mark)**

(ii)  $y = (x-2)^2 + 5$

$\therefore$  the inverse function is  $x = (y-2)^2 + 5$

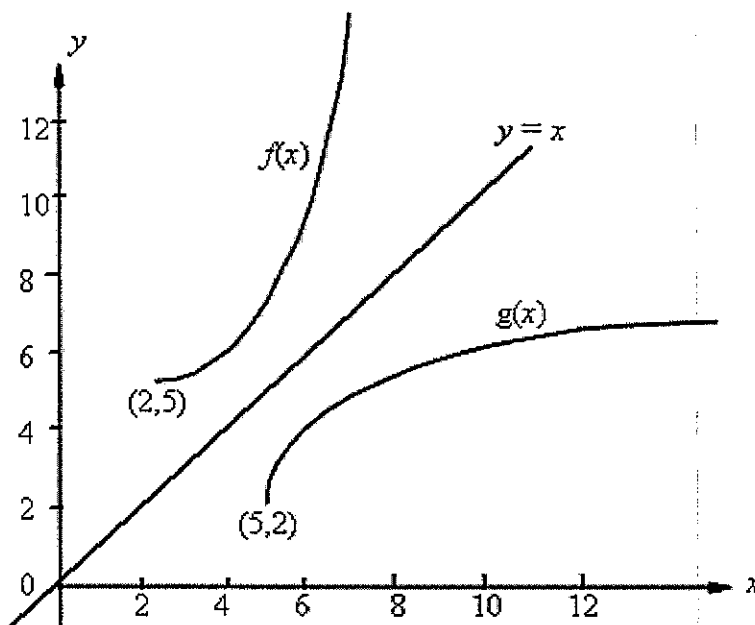
$$x - 5 = (y - 2)^2$$

$$\therefore y - 2 = \pm \sqrt{x - 5}$$

$$\therefore y = g(x) = 2 \pm \sqrt{x - 5} \quad \textbf{(1 mark)}$$

The domain of  $g(x)$  is  $x \geq 5$  and range is  $y \geq 2$ .

$$y = g(x) = 2 + \sqrt{x - 5}$$



**(1 mark each for the correct graphs of  $y = f(x)$  and  $y = g(x)$ ).**

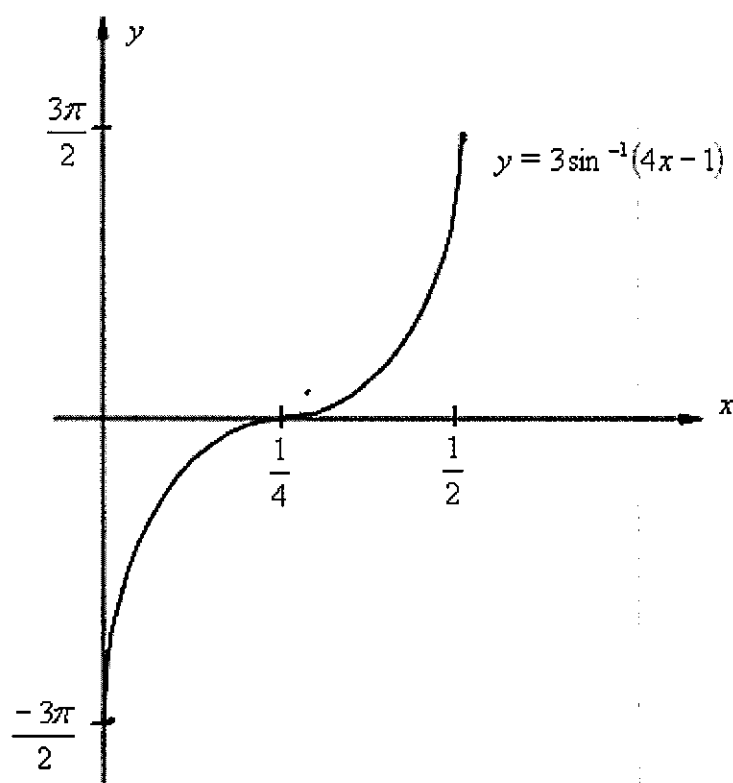
(b) (i) Domain:  $-1 \leq 4x - 1 \leq 1$   
 $0 \leq 4x \leq 2$   
 $0 \leq x \leq \frac{1}{2}$

**(1 mark)**

Range:  $-\frac{\pi}{2} \leq \frac{y}{3} \leq \frac{\pi}{2}$   
 $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$

**(1 mark)**

(ii)



( 1 mark for the graph)

(c)  $y = x \tan^{-1} x$

$$\frac{d(x \tan^{-1} x)}{dx} = x \cdot \frac{1}{1+x^2} + \tan^{-1} x \quad (1 \text{ mark})$$

$$\therefore \int_0^1 \tan^{-1} x \, dx = [x \tan^{-1} x]_0^1 - \int_0^1 \frac{x}{x^2+1} dx \quad (1 \text{ mark})$$

$$\begin{aligned} &= \frac{\pi}{4} - \frac{1}{2} [\log_e(1+x^2)]_0^1 \\ &= \frac{\pi}{4} - \frac{1}{2} \log_e 2 \quad (1 \text{ mark}) \end{aligned}$$

(d) (i)  $\int_0^{\frac{1}{2}} \frac{dx}{\sqrt{1-4x^2}} = \int_0^{\frac{1}{2}} \frac{dx}{\sqrt{4\left(\frac{1}{4}-x^2\right)}} = \frac{1}{2} \int_0^{\frac{1}{2}} \frac{dx}{\sqrt{\left(\frac{1}{4}-x^2\right)}} \quad (1 \text{ mark})$

$$= \frac{1}{2} \left[ \sin^{-1} \left( \frac{x}{\frac{1}{2}} \right) \right]_0^{\frac{1}{2}} \quad (1 \text{ mark})$$

$$= \frac{1}{2} \left( \frac{\pi}{2} - 0 \right) = \frac{\pi}{4} \quad (1 \text{ mark})$$

$$(ii) \int_0^{\frac{\pi}{2}} \frac{\cos x \, dx}{1 + \sin^2 x}$$

$$\text{let } u = \sin x$$

$$du = \cos x \, dx$$

$$\left. \begin{array}{l} \text{when } x = 0, \quad u = 0 \\ \text{when } x = \frac{\pi}{2}, \quad u = 1 \end{array} \right\} \quad (1 \text{ mark})$$

$$\therefore \int_0^{\frac{\pi}{2}} \frac{\cos x \, dx}{1 + \sin^2 x} = \int_0^1 \frac{du}{1 + u^2} = [\tan^{-1} u]_0^1 = \frac{\pi}{4} \quad (1 \text{ mark})$$

## Question 2

$$(a)(i) \quad \frac{d^2 x}{dt^2} = 9(x-2)$$

$$\frac{d\left(\frac{1}{2}v^2\right)}{dx} = 9(x-2)$$

$$\begin{aligned} \therefore \frac{1}{2}v^2 &= \int 9(x-2)dx \\ &= 9\left(\frac{x^2}{2} - 2x\right) + C \end{aligned}$$

When  $t = 0$ ,  $x = 4$  and  $v = -6$ .

$$\therefore 18 = 9(8 - 8) + C \quad \therefore C = 18 \quad (1 \text{ mark})$$

$$\therefore \frac{1}{2}v^2 = 9\left(\frac{x^2}{2} - 2x\right) + 18$$

$$\begin{aligned} \therefore v^2 &= 9x^2 - 36x + 36 \\ &= 9(x^2 - 4x + 4) \\ &= 9(x-2)^2 \text{ as required.} \quad (1 \text{ mark}) \end{aligned}$$

$$(ii) \quad v^2 = 9(x-2)^2$$

$$V = \pm 3(x-2)$$

$t = 0$ ,  $x = 4$  and  $v = -6$ .

$$\therefore v = -3(x-2) \quad (1 \text{ mark})$$

$$\text{So, } \frac{dx}{dt} = -3(x-2)$$

$$\therefore \frac{dt}{dx} = \frac{1}{-3(x-2)}$$

$$\begin{aligned} \therefore t &= \frac{-1}{3} \int \frac{1}{x-2} dx \\ &= \frac{-1}{3} \log_e(x-2) + C \end{aligned}$$

$$t = 0, \quad x = 4,$$

$$\therefore 0 = \frac{-1}{3} \log_e 2 + C$$

$$\therefore C = \frac{1}{3} \log_e 2$$

$$\therefore t = \frac{-1}{3} \log_e (x-2) + \frac{1}{3} \log_e 2$$

$$= \frac{1}{3} \log_e \frac{2}{x-2}$$

$$3t = \log_e \frac{2}{x-2}$$

$$\frac{2}{x-2} = e^{3t}$$

$$x-2 = 2e^{-3t}$$

$$x = 2(1 + e^{-3t}) \quad (1 \text{ mark})$$

(iii)  $v = -3(x-2)$

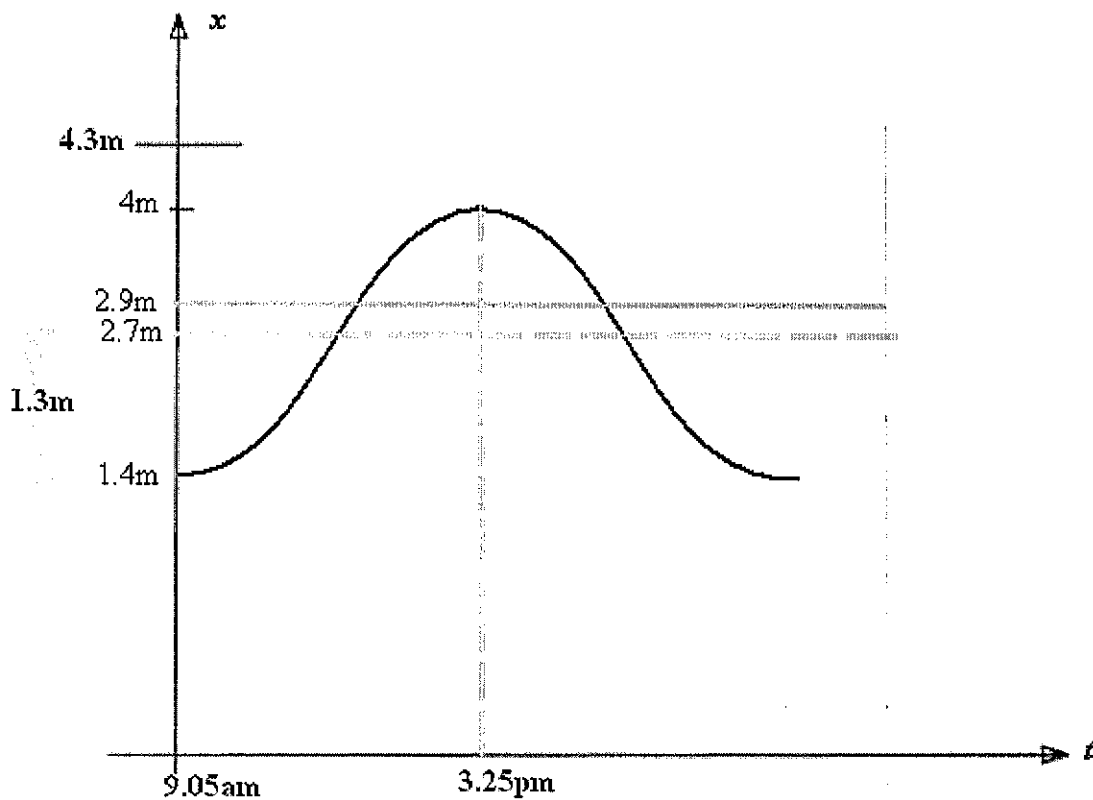
If  $v = 0$ ,  $x = 2$ . **(1 mark)**

We have  $x = 2(1 + e^{-3t})$

If  $x = 2$ ,  $2 = 2(1 + e^{-3t}) \rightarrow e^{-3t} = 0$  which has no solution.

$\therefore v$  is never zero. **(1 mark)**

(b)



**(1 mark for reasonable diagram)**

$$\text{Half-period} = 3.25 \text{ pm} - 9.05 \text{ am} = \frac{19}{3} \text{ hours}$$

$$\therefore \text{period } T = \frac{38}{3} \text{ hours}$$

$$\therefore n = \frac{2\pi}{T} = \frac{3\pi}{19}$$

$$\text{Displacement } x = 2.7 - 1.3 \cos\left(\frac{3\pi}{19}t\right) \quad \text{(1 mark for correct equation)}$$

The barge is 1.4 m above water level.

$\therefore$  the barge is level with the wharf when the water level is  $4.3 - 1.4 = 2.9 \text{ m}$

To find the time when water level is 2.9m,

$$x = 2.7 - 1.3 \cos\left(\frac{3\pi}{19}t\right)$$

$$2.9 = 2.7 - 1.3 \cos\left(\frac{3\pi}{19}t\right)$$

$$0.2 = -1.3 \cos\left(\frac{3\pi}{19}t\right)$$

$$\cos\left(\frac{3\pi}{19}t\right) = \frac{-0.2}{1.3}$$

$$\left(\frac{3\pi}{19}t\right) = 1.725\dots$$

$$t = 3.478\dots$$

$$= 3 \text{ h } 28 \text{ min } 41 \text{ sec}$$

$$= 3 \text{ h } 29 \text{ min}$$

$\therefore$  time = 3h 29 min + 9.05 am = 12.34pm. (1 mark correct time)

(from diagram there is only one solution between 9.05am and 3.25pm.)



(Solves for t: 1 mark)

### Question 3

$$(a) \quad (i) \log_e V = 3 \log_e (2R) - 0.81$$

$$0.81 = 3 \log_e (2R) - \log_e V$$

$$\log_e \left( \frac{(2R)^3}{V} \right) = 0.81 \quad \text{(1 mark)}$$

$$\frac{8R^3}{V} = e^{0.81}$$

$$\frac{8R^3}{V} = 2.25 \quad (\text{Given } e^{0.81} = 2.25) \quad \text{(1 mark)}$$

$$\frac{8R^3}{V} = \frac{9}{4}$$

$$V = \frac{4}{9} \times 8R^3$$

$$\text{ie. } V = \frac{32R^3}{9} \quad \text{(1 mark)}$$

$$(ii) \quad \frac{dR}{dt} = 0.002 \text{ m}^3/\text{year}$$

$$V = \frac{32R^3}{9}$$

$$\therefore \frac{dV}{dR} = \frac{32}{9} \times 3R^2 = \frac{32R^2}{3} \quad (1 \text{ mark})$$

$$\begin{aligned} \frac{dV}{dt} &= \frac{dV}{dR} \times \frac{dR}{dt} \\ &= \frac{32R^2}{3} \times 0.002 \end{aligned}$$

$$\begin{aligned} \text{When } R = 1.2, \quad \frac{dV}{dR} &= \frac{32 \times (1.2)^2}{3} \times 0.002 \quad (1 \text{ mark}) \\ &= 0.03072... \\ &= 0.031 \text{ m}^3/\text{year} \end{aligned}$$

$$\begin{aligned} (b) \quad \text{When } t = 0, T = 18 \\ \therefore 18 &= -19 + Ae^{-k \times 0} \\ \therefore A &= 19 + 18 = 37 \quad (1 \text{ mark}) \end{aligned}$$

$$\begin{aligned} (ii) \quad t = 5, T = 3 \\ 3 &= -19 + 37e^{-5k} \quad (1 \text{ mark}) \\ e^{-5k} &= \frac{22}{37} \\ k &= \frac{-1}{5} \log_e \frac{22}{37} \quad (1 \text{ mark}) \\ &= 0.103975091 \end{aligned}$$

$$\begin{aligned} \text{When } T = -18.9, \\ -18.9 &= -19 + 37e^{-kt} \\ 37e^{-kt} &= 0.1 \\ e^{-kt} &= \frac{0.1}{37} \end{aligned}$$

$$-kt = \log_e \left( \frac{0.1}{37} \right)$$

$$\therefore t = \frac{\log_e \left( \frac{0.1}{37} \right)}{-k} = 56.8742...$$

$$\therefore t = 57 \text{ minutes.} \quad (1 \text{ mark})$$

Question 4

Now,  $36\text{km/h} = \frac{36000}{60 \times 60} = 10\text{m/s}$

Also, when  $t = 0$ ,  $x = y = 0$ ;  $\dot{x} = 10\text{m/s}$  and  $\dot{y} = 0$  (clouds are moving horizontally)

(i) Horizontally

$$\ddot{x} = 0$$

$$\therefore \dot{x} = C_1$$

$$t = 0, \dot{x} = 10 \rightarrow C_1 = 10$$

$$\therefore \dot{x} = 10$$

$$x = \int 10 dt = 10t + C_2$$

$$t = 0, x = 0 \rightarrow C_2 = 0$$

$$x = 10t$$

Vertically

$$\ddot{y} = -g = -10 \text{ ms}^{-2}$$

$$\dot{y} = \int -10 dt$$

$$= -10t + C_3$$

$$t = 0, \dot{y} = 0 \rightarrow C_3 = 0$$

$$\dot{y} = -10t$$

$$y = -5t^2 + C_4$$

$$t = 0, y = 0 \rightarrow C_4 = 0$$

$$y = -5t^2$$

**(1 mark each for correct derivation of equations of motion)**

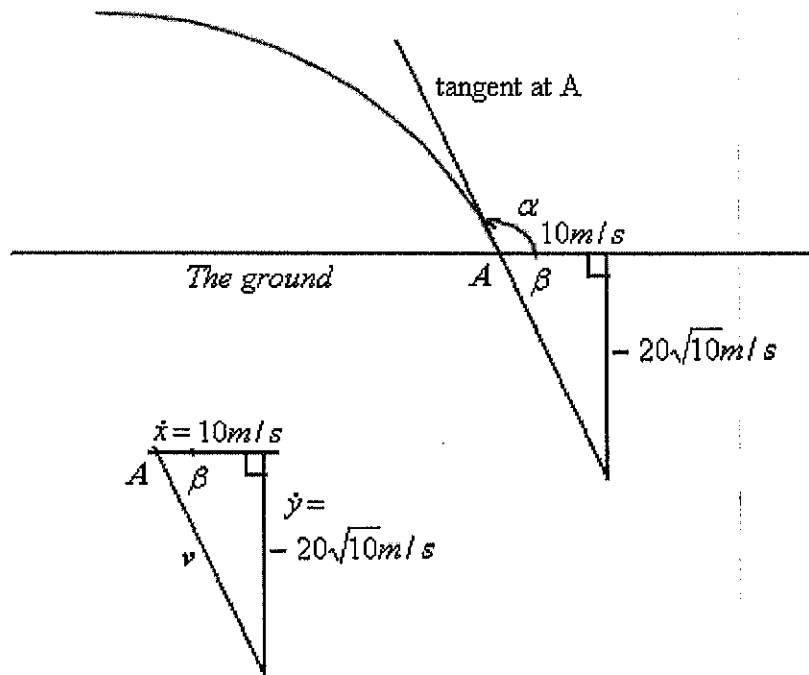
(ii) The raindrop falls 200m. ie.  $y = -200$

$$-5t^2 = -200$$

$$t^2 = 40$$

$$t = 2\sqrt{10} \text{ seconds } (t > 0)$$

(iii)



**(velocity Diagram 1 mark)**

When the raindrop hits the ground (at A say),  $t = 2\sqrt{10}$  seconds.

Then  $\therefore \dot{x} = 10 \text{ m/s}$  in the  $\rightarrow$  direction and

$$\dot{y} = -10t = -10 \times 2\sqrt{10} = -20\sqrt{10}\text{m/s} \text{ in the } \downarrow \text{ direction.}$$

$$|v| = \sqrt{\dot{x}^2 + \dot{y}^2} = \sqrt{(10)^2 + (-20\sqrt{10})^2} = 10\sqrt{41} \text{ m/s } \textbf{(1 mark)}$$

$$\text{Also, } \tan \beta = \frac{|\dot{y}|}{\dot{x}} = \frac{|-20\sqrt{10}|}{10} = 2\sqrt{10}$$

$$\therefore \beta = 81^\circ 1'$$

$$\alpha = 180 - \beta = 180 - 81^\circ 1' = 98^\circ 59'$$

The raindrop hits the ground at an angle of  $98^\circ 59'$ . **(1 mark)**

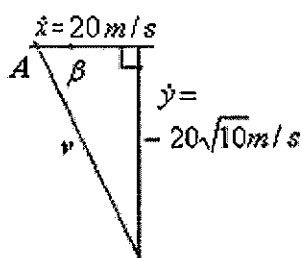
(iv) When the wind speed doubles,  $\dot{x} = 20$  m/s.

Using the initial conditions, the raindrop still falls 200m and therefore the time

Taken to fall to the ground, still remains as  $2\sqrt{10}$  seconds.

$$\therefore \dot{y} = -10t = -10 \times 2\sqrt{10} = -20\sqrt{10} \text{ m/s}$$

**(1 mark)**



$$\text{From the velocity diagram, } \tan \beta = \frac{|\dot{y}|}{\dot{x}} = \frac{|-20\sqrt{10}|}{20} = \sqrt{10}$$

$$\therefore \beta = 72^\circ 27'$$

$$\alpha = 180 - \beta = 180 - 72^\circ 27' = 107^\circ 33'$$

The raindrop hits the ground at an angle of  $107^\circ 33'$ . **(1 mark)**